The Geometry of Learning: Data Complexity Controls the Fractal Dimension of Gradient Descent

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We measure the fractal dimension of neural network training trajectories in function space. Using the Grassberger-Procaccia correlation dimension D_2, we find that a convolutional network trained on CIFAR-10 with full-batch gradient descent produces a strange attractor with $D_2 = 3.67 \pm 0.08$ at 30% of the Edge of Stability threshold. Persistent homology confirms fractal rather than smooth manifold topology. Cross-architecture controls reveal that data complexity is the necessary ingredient for multi-dimensional chaos, while architecture modulates the learning-rate threshold at which it appears: a multilayer perceptron on the same data reaches $D_2 = 4.37 \pm 0.05$ near the Edge of Stability, whereas either architecture on structureless synthetic data remains at $D_2 \approx 0.9$. A continuous label-noise sweep confirms that data structure controls attractor complexity, but reveals a regime dependence: at moderate learning rates, degrading label structure smoothly reduces D_2 from 3.6 to 1.7; near the Edge of Stability, the rougher landscape created by random labels slightly increases D_2. Lyapunov exponents peak at intermediate learning rates and become negative at high learning rates as basin convergence dominates. Peak chaos and peak geometric complexity dissociate across all architectures tested—the attractor is most complex in the transition zone between chaotic and basin-convergent regimes, consistent with the KAM transition. These results provide a geometric characterization of training dynamics consistent with the Newhouse-Ruelle-Takens route to chaos.

INTRODUCTION

Gradient descent is a deterministic dynamical system. Each update maps the network's function to a new function, producing a trajectory through function space whose geometry can be measured.

Recent work has established that this geometry is nontrivial. Cohen *et al.* [1] discovered the Edge of Stability (EoS): the top Hessian eigenvalue rises to $2/\eta$ and oscillates rather than diverging. Damian *et al.* [2] characterized the self-stabilization mechanism. Kalra *et al.* [3] and Ghosh *et al.* [4] observed period-doubling near the EoS, and Morales *et al.* [5] computed Lyapunov exponents during training, establishing that the dynamics can be chaotic. Storm *et al.* [6] showed that finite-time Lyapunov exponents through network depth form coherent structures in input space.

These results establish that training dynamics are rich, but leave open a geometric question: what *shape* does the chaos have? A chaotic trajectory might be essentially one-dimensional—erratic motion along a single convergence path—or it might fill a multi-dimensional fractal volume. We measure this distinction using the correlation dimension D_2 [7] and find that training trajectories settle onto strange attractors whose fractal dimension is controlled by data complexity and modulated by architecture.

EXPERIMENTAL FRAMEWORK

Architectures and data.—We study five conditions under full-batch gradient descent with MSE loss, no momentum, and no weight decay (Table I). Full-batch training isolates the deterministic dynamics. Learning rates are expressed as fractions of $2/\lambda_{\max}$, where $\lambda_{\max}$ is the top Hessian eigenvalue after 1,000 warmup steps at $\eta = 0.01$. All networks are trained for 5,000 steps.

The CNN has two convolutional layers (3→16→32 channels, 3×3 kernels, max pooling, ReLU, two fully connected layers; 268,650 parameters) trained on a 2,000-image CIFAR-10 subset. The MLPs use 2-hidden-layer tanh networks of width 50 (156K parameters) or 85 (269K parameters) on either CIFAR-10 (flattened) or synthetic data (220 dimensions, 10 classes). The CNN-on-synthetic condition embeds synthetic data into 3×32×32 tensors to isolate architectural effects.

Lyapunov exponents.—We measure chaos in function space to avoid overparameterization artifacts. Two copies of the network train from identical initialization, one perturbed by $\varepsilon = 10^{-5}$ along a random unit-norm direction. The function-space divergence $\delta(t) = \|f(\mathbf{x}) - \tilde{f}(\mathbf{x})\|$ is recorded on 100 held-out inputs; λ is the slope of $\ln \delta(t)$ versus t over $[0.2T, 0.8T]$, with t measured in training steps so that λ has units of inverse steps. Sensitivity analysis confirms that $\varepsilon = 10^{-8}$ produces uniform artifactual positive exponents; $\varepsilon \geq 10^{-6}$ is required for converged measurements. The choices $\varepsilon = 10^{-5}$ and 100 held-out inputs are within a plateau verified in the supplement [8]. Results are averaged over 10 seeds (CNN/CIFAR-10, MLP/CIFAR-10, CNN/synthetic) or 20 seeds (MLP/synthetic).

Correlation dimension.—D_2 is computed via the Grassberger-Procaccia algorithm [7] on function-space trajectories (outputs recorded every 10 steps, first 20% discarded as transient, yielding ∼400 points). Pipeline

validation recovers $D_2 = 2.18$ for a quasiperiodic 2-torus (expected 2.0) and $D_2 = 1.73$ for the Lorenz attractor (expected 2.05). Extended calibration [8] reveals a systematic 15–35% underestimate for fractal attractors; a $D_2(N)$ convergence analysis confirms that reported values are conservative lower bounds that have not yet saturated at $N{\approx}400$ [8].

Persistent homology.—Computed via RIPSER [9] on PCA-reduced trajectories to distinguish smooth tori (few persistent loops, large gap ratio) from strange attractors (many features at all scales, gap ratio ≈1).

RESULTS

The chaos window.—The Lyapunov exponent is non-monotonic with learning rate [Fig. 1(a)]. In the CNN on CIFAR-10, λ peaks at 15% of the EoS threshold and crosses zero near 45%. Above this, basin convergence dominates: trajectories from different perturbations collapse to the same learned function. This bounded chaos window appears in every architecture tested; its width scales with architecture complexity.

Training trajectories are strange attractors.—Within the chaos window, the CNN/CIFAR-10 trajectory acquires fractal structure [Fig. 1(b)]. D_2 crosses 1.0 at 10% of EoS, rises through 2.63 ± 0.62 at 20%, peaks at 3.67 ± 0.08 at 30%, and remains above 2.0 to 80% of EoS. The first principal component captures only ∼52% of trajectory variance—half the dynamics are off the primary convergence axis. Since our pipeline underestimates fractal dimensions by 15–35%, the true attractor dimension is likely substantially higher.

Persistent homology confirms fractal topology across architectures [8]. For the CNN at 30% of EoS, 386 H_1 features appear with gap ratio 1.2 (no dominant loops); at 5% of EoS, zero H_1 features: a simple convergence path. Both MLPs on CIFAR-10 show the same transition: zero H_1 features below 30% of EoS, rising to 389 (H_1) and 337 (H_2) features at 90% for the 156K MLP, comparable to the CNN at its peak. The gap ratio remains near 1.0 in every condition—the topological signature of a strange attractor, not a smooth torus.

Data complexity is the engine; architecture is the throttle.—The cross-architecture controls (Table I, Fig. 3) reveal a clean hierarchy. A CNN on structureless synthetic data produces $D_2 < 1.0$ at all learning rates—architecture alone is insufficient. A parameter-matched MLP (269K) on CIFAR-10 reaches $D_2 = 4.37 \pm 0.05$ at 90% of EoS; a smaller MLP (156K) reaches $D_2 = 4.61 \pm 0.10$. Both sit near the Eckmann–Ruelle ceiling for our trajectory length, but the width inversion is significant at 2.2σ: the wider network's additional capacity raises the effective stability margin, delaying bifurcation onset and slightly suppressing peak D_2 at a given learning-rate fraction. Width modulates the pace of the transition to chaos, not its destination. Data complexity is both necessary (CNN + synthetic → $D_2 \approx 1$) and sufficient (MLP + CIFAR-10 → $D_2 > 4$).

Architecture modulates the *threshold*: at 30% of EoS the CNN has already reached $D_2 = 3.67$ while both MLPs remain at $D_2 \approx 1$. The MLPs require 70–90% of EoS to achieve comparable fractal dimension. The convolutional hierarchy facilitates mode coupling at lower learning rates.

Data complexity as a continuous control parameter.—To move beyond the binary CIFAR/synthetic comparison, we sweep label-noise fraction p from 0 (clean CIFAR-10) to 1 (uniformly random labels) at each architecture's peak-D_2 learning rate (Fig. 2). For the CNN at 30% of EoS, D_2 falls smoothly from 3.64 ± 0.11 to 1.68 ± 0.54—randomizing labels degrades data structure and destroys the attractor modes. However, for the MLP at 90% of EoS, D_2 rises modestly from 4.36 ± 0.05 to 4.77 ± 0.08. This divergence reflects a regime dependence: at moderate learning rates, structured data creates the oscillatory modes that the attractor fills, and noise removes them; near the Edge of Stability, random labels roughen the loss surface, creating additional competing minima that sustain higher-dimensional dynamics even without label structure. In both regimes, changing label content continuously changes the attractor—data complexity acts as a genuine control parameter, not a binary switch.

Peak chaos and peak complexity dissociate.—For the CNN on CIFAR-10, peak λ occurs at 15% of EoS while peak D_2 appears at 30%. The same dissociation is universal across architectures: the MLP (156K) peaks in λ at 40% versus D_2 at 90% of EoS; the MLP (269K) at 50% versus 90%. In every case the attractor is geometrically most complex not where perturbation sensitivity is greatest, but in the transition zone where chaotic dynamics and basin convergence compete. This cross-architecture dissociation is characteristic of the KAM transition regime [10], where ordered and chaotic regions coexist in fractal interleaving.

TABLE I. Experimental conditions with D_2 at each condition's peak.

	Params	Data	Peak D_2	EoS%	N
MLP/synth.	14K	synth.	0.9	all	20
CNN/synth.	269K	synth.	0.98(3)	90	10
MLP/CIFAR	156K	C-10	4.61(10)	90	10
MLP/CIFAR	269K	C-10	4.37(5)	90	10
CNN/CIFAR	269K	C-10	3.67(8)	30	10

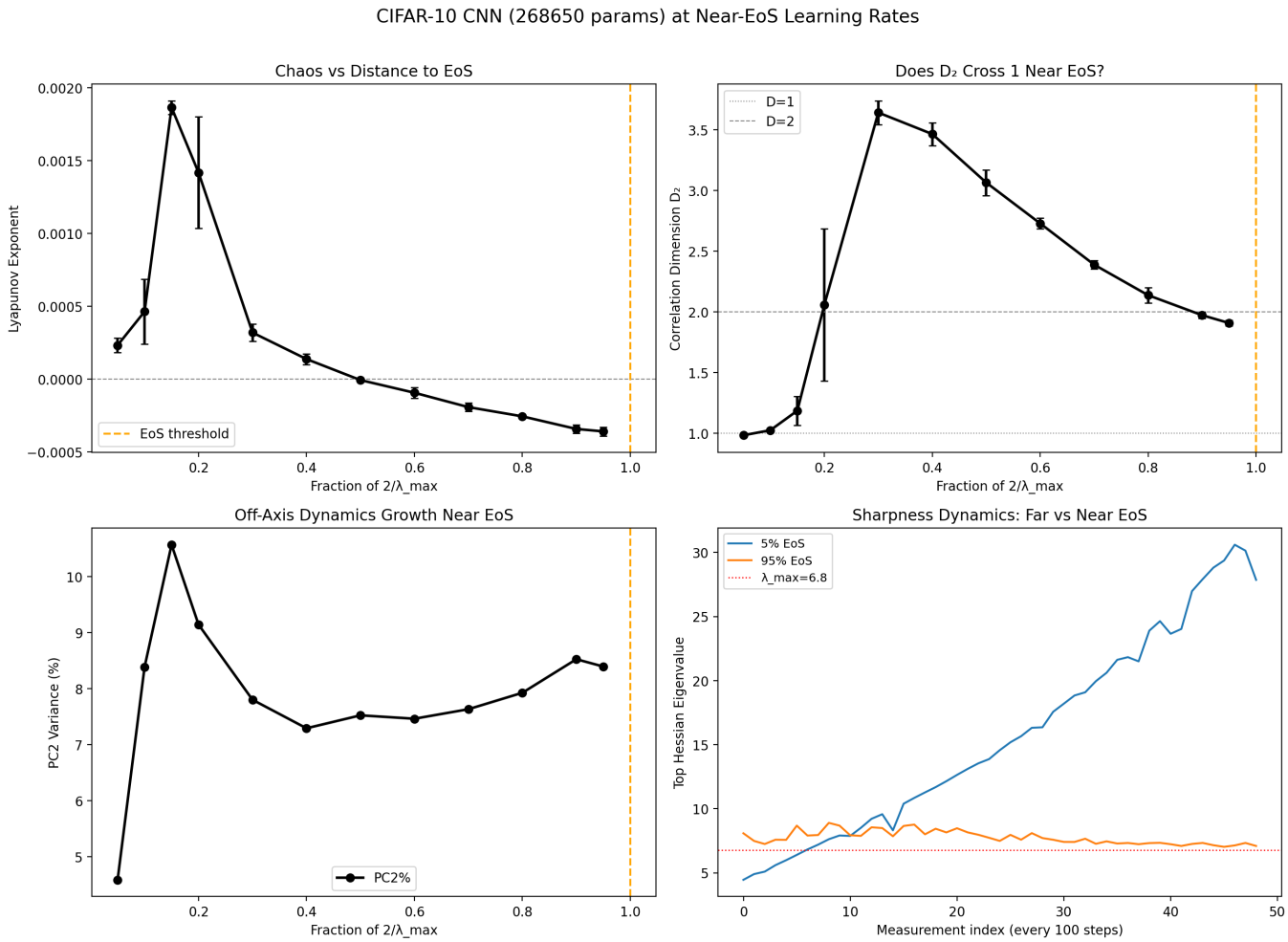

FIG. 1. CNN (268,650 parameters) on CIFAR-10, 10 seeds. (a) Lyapunov exponent versus fraction of the EoS threshold $2/\lambda_{\max}$. The chaos window peaks at 15% and closes near 45%. (b) Correlation dimension D_2. The trajectory crosses $D_2 = 1$ at 10% of EoS and peaks at 3.67 ± 0.08 at 30%. (c) Second principal component variance, showing off-axis dynamics. (d) Sharpness time series: monotonic progressive sharpening at 5% EoS (blue) versus bounded oscillation at 95% EoS (orange). Error bars: standard deviation across seeds.

DISCUSSION

These results provide a geometric characterization of gradient descent: the trajectory through function space is a strange attractor whose fractal dimension depends on what the network is learning and how it is structured.

The findings are consistent with the Newhouse-Ruelle-Takens route to chaos [11, 12]. Complex data creates a loss landscape with ridges, saddle points, and near-equivalent solutions that provide independent axes of oscillation. When enough of these modes couple through the training dynamics, the flow produces a strange attractor. The label-noise sweep (Fig. 2) reveals that data complexity acts as a continuous control parameter, but its effect is regime-dependent: at moderate learning rates, structured labels create the oscillatory modes that the attractor fills, and degrading them smoothly reduces D_2; near the Edge of Stability, even random labels sustain high-dimensional chaos because the rougher loss surface provides its own competing modes. The bounded chaos window is consistent with the KAM picture [10]: some invariant structures survive perturbation while others are destroyed. The universality of the λ–D_2 dissociation across architectures—each showing peak geometric complexity offset from peak sensitivity—strengthens this identification: the transition zone where chaos and order coexist produces the richest attractor geometry regardless of the network that mediates the dynamics. The CNN's hierarchical processing creates efficient coupling pathways, lowering the learning rate at which the transition occurs. The MLP accesses the same modes—provided by the data—but requires stronger driving to couple them. On structureless data, no architecture produces multi-dimensional dynamics because the modes do not exist.

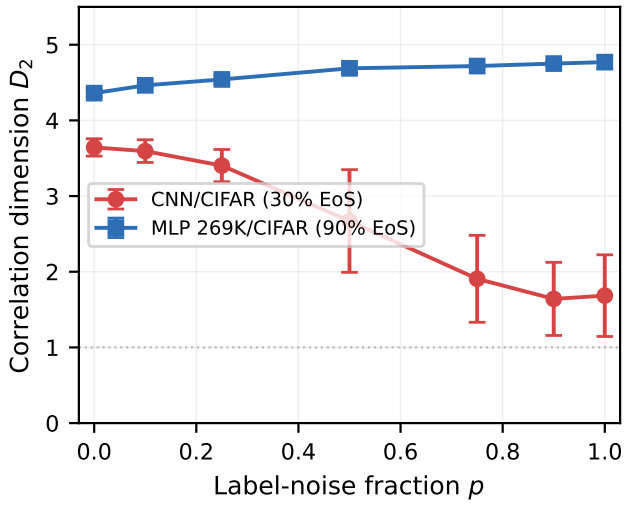

FIG. 2. D_2 versus label-noise fraction p at each architecture's peak learning rate. CNN at 30% EoS (red circles): D_2 falls as label structure is degraded. MLP 269K at 90% EoS (blue squares): D_2 rises modestly as random labels roughen the loss landscape. Three seeds; error bars: standard deviation.

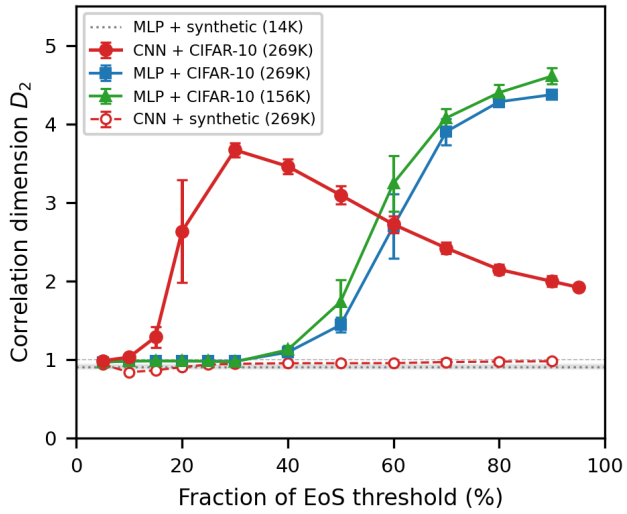

FIG. 3. D_2 versus fraction of the EoS threshold for all five conditions. CNN on CIFAR-10 (red circles) peaks at 30% EoS; MLPs on CIFAR-10 (blue squares, green triangles) peak at 90% EoS. Synthetic conditions (open markers) remain at $D_2 \approx 1$ throughout. All CIFAR and CNN/synthetic conditions: 10 seeds; MLP/synthetic: 20 seeds. Error bars: standard deviation across seeds.

These dynamical findings complement Pittorino *et al.* [13], who showed that the loss landscape has toroidal topology after symmetry removal—a static geometric property of the parameter space on which these dynamical transitions unfold.

A methodological note: Lyapunov exponents computed at perturbation scale $\varepsilon = 10^{-8}$ are dominated by numerical noise [8]. Reliable measurements require $\varepsilon \geq 10^{-6}$. Prior work at smaller ε [5] should be interpreted with caution. The function-space protocol avoids false positives from gauge symmetries in overparameterized networks.

Several questions remain open. Does the attractor structure persist under stochastic gradient descent? Does D_2 continue to grow with larger architectures and datasets? Networks trained within the chaos window are known to generalize better [1], and our measurements show these are the learning rates that produce strange attractors. Whether this connection is causal—whether exploring a higher-dimensional attractor during training produces better solutions—is the central question these measurements now enable.

AI language models (Claude, Anthropic) were used as research tools throughout this work, including assistance with experimental design, code development, data analysis, and manuscript preparation. All research directions, experimental choices, and scientific conclusions are the sole responsibility of the author. Source code, raw data, and analysis scripts are available at `https://github.com/gnomevan/attractor`.

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[8] See Supplemental Material at [URL will be inserted by publisher] for perturbation sensitivity analysis, D_2 pipeline calibration on known attractors (Hénon, Lorenz, Rössler, Mackey-Glass, smooth tori), extended D_2 values at all learning rate fractions, persistent homology

details for all architectures, D_2 convergence with trajectory length, convergence of D_2 and λ with measurement parameters, depth scaling, label-noise sweep data, and λ–D_2 dissociation analysis.

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